ITU • AI FOR GOOD SANDBOX HACKATHON

SADA AI

THE WARNING THAT NEVER REACHED

Policy-grounded, multi-agent emergency warning resilience planning

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ASTRA

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CODE MERGED • 531 BACKEND TESTS PASS • LIVE TEST DEPLOYMENT • AI REVIEW GATES

01 Introduction

An emergency warning may be sent yet fail to arrive on time, remain inaccessible, or lack actionable instructions on the available channel. SADA AI models these last-mile gaps BEFORE emergencies. Using fixed synthetic inputs, it compares a fixed-priority baseline (P0) with feasible, accessibility-aware recovery plans (P1–P3).

Four coordinated rule-based agents simulate delivery, assess risk, review policy/accessibility and compare recovery plans. Separate evidence-grounded Qwen explanations support human review; AI-generated text cannot change simulated delivery or KPIs. This is a pre-event planning prototype, not a live warning service, safety guarantee or legal certification tool.

02 Description of the use case

A planner selects a cumulative snapshot (T+00 / T+02 / T+04 / T+06) and at least one of five approved channels: CH01 SMS, CH02 mobile app, CH03 automated voice, CH04 public siren, CH05 digital signage. The canonical warning remains fixed in Arabic; CH04 sirens are alert-only and cannot establish effective actionable coverage by themselves.

1,000	05	04
SYNTHETIC RESIDENTS	DELIVERY CHANNELS	EVENT SNAPSHOTS

Three zones: A 400 • B 350 • C 250. Profile flags (overlapping groups): 65+ 150; 60+ 228; hearing 80; visual 60; limited internet 250; weak coverage 120; app installed 600. EV01 interrupts SMS for 245 Zone B residents (T+02 to T+07). EV02 activates 30 voice and 20 signage barriers at T+04: 50 resident–channel cases affecting 48 people. Original dataset remains unchanged.

HOW A PLAN IS EVALUATED

01	02	03	04
DELIVERY	**RISK**	**POLICY + ACCESS**	**RECOVERY**
Deterministic attempts	Failed/at-risk groups	Usability / review	Compare P0–P3

KPIs: Estimated Reach; Effective Warning Coverage (arrived + usable + actionable); Recovery Improvement; Unresolved Critical Cases. P0 order: SMS → Voice → App → Siren → Signage. Delivered ≤30 s; Delayed >30–60 s; Failed/Unavailable distinct. App installation and siren/signage area are hard prerequisites; weak coverage/internet is a risk, not exclusion. No Eligible Channel is flagged (no P0 automatic backup). All estimates are synthetic.

03 Use case requirements

IMPLEMENTATION REQUIREMENTS • BOUNDARIES • VERIFICATION

Requirements map to prototype evidence. Priority: C = critical, E = expected (team triage, not an ITU scoring rule); no R1–R6 item is classified optional.

NODE	TESTABLE REQUIREMENT	IMPLEMENTATION / AUDIT EVIDENCE
R1 DATA (C)	Read 1,000 distinct synthetic resident profiles and retain zone, connectivity, disability and channel-prerequisite flags.	Approved Step 1 master workbook; validations; fixed timeline; no live resident data.
R2 STAGES (C)	Evaluate cumulative T+00/T+02/T+04/T+06 evidence; preserve permanent restrictions and do not invent early barrier origins.	EV02 activation at T+04; pre-onset unproven state remains Unknown; 50 cases × 4 snapshots.
R3 ENGINE (C)	Deterministic Delivered (≤30 s), Delayed (>30 to ≤60 s), Failed and Unavailable states; only eligible/selected channels.	P0 fixed priority; accessibility-aware P1–P3 reruns under identical snapshots; siren alert-only.
R4 KB/AI (E)	Retrieve attributable source excerpts and flag unsupported, contradictory or insufficient-evidence AI claims.	Runtime KB: 5 sources/11 excerpts with lexical retrieval; Qwen explains but never sets KPIs. New five-source/15-summary UAE reference JSON is documentation-only and not indexed.
R5 OUTPUT (C)	Expose per-channel results, before/after gaps, plan trade-offs, citations and unresolved reviews.	HTML dashboard and Markdown report; React/Node/FastAPI source merged into GitHub. Public test deployment is separate from code merge.
R6 PRIVACY (C)	Protect source provenance; keep warnings fixed and simulation synthetic; route consequential issues to humans.	No live personal-data processing in demo; real deployment needs authority, security and privacy review.

TWO ISOLATED PROCESSING PATHS

DETERMINISTIC PLANNING ENGINE

Agent 1 Delivery → Agent 2 Risk → Agent 3 Policy & Accessibility → Agent 4 Recovery. Reruns compare P0 against P1–P3. AI text cannot alter delivery states, plan ranking, or KPIs.

GROUNDED EXPLANATION LAYER

Question → approved lexical evidence retrieval → Qwen explanation (local Ollama 3B or Replit/OpenRouter 7B) → grounding/citation checks → answer or ReviewRequired → human review. No CAP certification or legal verdict.

RELEASE AND REPRODUCIBILITY

06 Nov 2026: GitHub PR #5 merged the website sources into main (merge 401aff4). All 3 CI checks passed; SADA Linux backend: 531 passed, 10 skipped, 795 subtests passed. The public Replit test deployment reports Success. Deployment-specific security, code-version parity, frontend checks and accepted AI answers remain unverified.

```
python -m unittest discover -s tests -p "test_*.py"
```

```
python demo/run_ethar_stage_demo.py --stage T+04 --output-dir demo_ethar_T04
```

04

ITU-T Y.3172 • AI readiness • policy mapping

FUNCTIONAL ANALOGY, NOT STANDARDS CONFORMITY

Y.3172 provides architectural analogy, not certification. The nodes describe actual synthetic inputs, handling and outputs; no real resident data were collected or de-identified. Readiness lenses are selected, not scored [2,3].

Y.3172 NODE	SADA AI IMPLEMENTATION	READINESS LENS	POLICY / EVIDENCE
SRC Source	Fixed 1,000 synthetic profiles, 5 channels; 11 KB excerpts.	Data quality / provenance	Fixed dataset; CAP 1.2 [4]
C Collector	Read-only synthetic workbook; lexical KB read.	Data service / infrastructure	PDPL review context [6]
PP Pre-process	Validate schema/EV01–EV03; no de-ID.	Data governance / interoperability	Accessibility policy [5]
M Model	Fixed P0–P3 reruns; Qwen cannot set KPIs.	AI assurance / automation	Y.3172; no KPI edits [2]
P Policy	Agent 3: usability, evidence, privacy review.	Human oversight / AI policy	CAP, UAE policy, PDPL [4–6]
D Distributor	HTML/Markdown metrics, reasons and source IDs.	Human interface / accountability	Citations and review flags
SINK Decision	Human decision; never sends live alerts.	Inclusion / collaboration	Human approval; no dispatch

READINESS 2.0: PROJECT-LEVEL INTERPRETATION

DATA	1,000 synthetic profiles and a 5-source/11-excerpt lexical runtime KB; 5 supplementary UAE references are documentation-only, not indexed.
DIGITAL INFRASTRUCTURE	EV01 disruption and weak-coverage stress; local model proof; no real-network, cloud-capacity or energy audit.
SKILLS / INNOVATION	Tests and source traceability; no planner training study or ITU-hosted vector knowledge base.
AI POLICY	CAP/UAE privacy, accessibility and AI-governance references; human review is not an automated approval workflow.
INCLUSION / HUMAN INTERFACE	Disability-aware stage modeling, fixed Arabic warning, traceable recommendations; end-user accessibility research pending.

POLICY SCOPE AND BOUNDARIES

CAP 1.2 [4] · An optional instruction field does not prove actionable or accessible delivery.

UAE ACCESSIBILITY [5] · Guides inclusive digital-service design; emergency-channel applicability requires official review.

UAE PDPL [6] · Synthetic demo only; any future identifiable disability profiling/sharing demands legal and privacy assessment.

UAE-ALERT [7] · Cell Broadcast specifications are NOT CH01 SMS; no 3GPP or Cell Broadcast simulation.

Open: source recall, Arabic access, stakeholder review and deployment security/energy. Five UAE policy references are GitHub documentation only, not runtime indexed; no ITU vector KB.

05

Error scenarios & resilience evaluation

PROGRESSIVELY REVEALED TECHNICAL AND GOVERNANCE STRESSES

Stage selection reveals evidence cumulatively while candidate plans are retested under matching conditions. Permanent restrictions persist; unproven EV02 barriers remain Unknown before T+04, then Active from T+04. The policy-misuse scenario illustrates a human review requirement; it is not proof of access-control enforcement or penetration testing.

STAGE	INJECTED SCENARIO	AGENT RESPONSE / HUMAN DECISION
T+00 BASELINE	Fixed Arabic warning; five channels; synthetic residents.	Evaluate fixed P0 and accessibility-aware alternatives; record risks/gaps.
T+02 EV01	CH01 SMS outage affects 245 residents in Zone B until T+07.	A1 detects failures → A2 flags risk → A4 compares viable reruns.
T+04 EV02	30 voice + 20 signage resident–channel barriers (48 unique people).	A3 distinguishes permanent / Unknown / active EV02; A4 avoids inaccessible paths.
T+06 EV03	EV01/EV02 persist; EV03 adds hypothetical external-route policy review.	A3 flags Authorization/Privacy review; A4 reruns only internal plans.
POLICY CHALLENGE	Attempted disclosure of identifiable risk profiles or blanket legal/CAP claim.	Unsupported legal/CAP conclusions require review. Restricting sensitive-profile export is a proposed control, not demonstrated by this scenario.

MEASURED SYNTHETIC OUTCOME • T+02 • FIVE CHANNELS

ESTIMATED REACH	EFFECTIVE COVERAGE	UNRESOLVED CASES
722 → **942**	0 → **366**	278 → **58**
residents reached	actionable + usable	technical cases

Recommended plan: P3. Modeled reach increases 722 → 942; effective usable/actionable warning coverage rises 0 → 366; unresolved technical cases decrease 278 → 58. Outputs are simulation estimates, NOT a measurement of real residents' understanding or protection.

IMPORTANT TRADE-OFF

At T+00, selecting P3 improves effective coverage but lowers technical reach from 961 to 946. The model explicitly surfaces this rather than hiding the cost.

SCENARIO ROBUSTNESS

T+04 shows the same aggregate reach and effective-coverage totals but different EV02 classifications. At T+02 with CH01 disabled, P3 reach is 947 with 53 unresolved technical cases.

OBSERVED EVIDENCE VS OPEN ACCEPTANCE GATES

PASSED / OBSERVED: PR #5 merged; 3 GitHub CI checks green; Linux backend 531 passed, 10 skipped, 795 subtests; local simulation outcomes and 11 excerpts / five sources. Local Gitleaks: 43 commits, no leaks detected; Dependabot: 0 open alerts (06 Nov). Two real English/Arabic OpenRouter AI requests through Replit Preview returned ReviewRequired, not displayed explanatory answers.

OPEN VERIFICATION: Replit reports the website test deployment published successfully, but live deployment parity/end-to-end checks, frontend build/tests, UI accessibility and host security validation are not independently complete. Replit source may have later changes than the GitHub import; do not present the two Preview AI tests as production-deployment acceptance tests.

06

Policy & standards gaps

GAPS SURFACED BY THE MAPPING EXERCISE

GAP	POLICY LEARNING VALUE / PROPOSED MITIGATION
G1 REACH ≠ ACTION	Separate arrival from actionable warnings; compare with TDRA telecom continuity planning and UN CRPD accessibility principles [8].
G2 EVIDENCE / TRACE	Trace input → Agent output → reason/source → review status; OECD AI Principles inform transparency.
G3 CHANNEL SCOPE	SMS is not UAE-Alert Cell Broadcast; no live operator integration. TDRA continuity work remains future scope.
G4 AUTHORITY / PRIVACY	Decree-Law 8/2025 informs institutional roles; UAE PDPL is local context; GDPR Article 5 is comparative [6,10].
G5 AI / KB GOVERNANCE	UAE AI Strategy 2031, AI Cybersecurity Policy, UAE AI Charter and NIST AI RMF guide gaps; new sources are not indexed.
G6 DEPLOYMENT GATES	WCAG 2.2, Arabic and screen-reader checks plus security, energy and IP review remain future pilot gates [9].

EVIDENCE & DELIVERY STATUS

VALIDATED BACKEND

06 Nov 2026: PR #5 merged at 401aff4, 3/3 CI checks passed; Linux backend 531 passed, 10 skipped, 795 subtests. Earlier local 255/255 tests. Gitleaks reviewed 43 local commits (0 findings); Dependabot showed 0 open alerts. These checks do not establish full host security.

REMAINING FOR SUBMISSION

Replit test deployment needs code-parity, UI accessibility and host-security checks. Two bilingual AI Preview calls returned ReviewRequired, not accepted answers. The GitHub 5-source/15-summary UAE register is documentation only. Keep screenshots and correct submission links.

Code paths: website/artifacts/sada/ (React), website/artifacts/api-server/ (Node), website/backend/ (Python/FastAPI); root lastmile_ai/ (deterministic engine, retrieval and AI review). Synthetic master: data/reference/LastMile_AI_Step1_Master_Dataset.xlsx. Nominal scenario assumptions: SMS 90%, app 70%, voice 85%, siren 75%, signage 40%.

AUTHORITATIVE EVIDENCE REGISTER

[1] ITU AI for Good Sandbox Hackathon • UAE

[2] ITU-T Y.3172 (2019) • ML architectural framework

[3] ITU AI Ready • Standardized Readiness Framework

[4] OASIS Common Alerting Protocol 1.2 • §3.2

[5] UAE National Digital Accessibility Policy

[6] UAE Federal Decree-Law No. 45/2021 • PDPL

[7] UAE-Alert TS FS 001 • Cell Broadcast

[8] UN CRPD Articles 9 / 11 • accessibility/emergencies

[9] W3C Web Content Accessibility Guidelines 2.2

[10] GDPR Regulation 2016/679 • Art. 5 comparator

REPOSITORY AND SCOPE DECLARATION

PROJECT REPOSITORY :https://github.com/LastMileProject0/Sada
LIVE SADA WEBSITE: https://safe-integration-testing-copy-of-sada.replit.app

SADA AI is a synthetic pre-event decision-support prototype. GitHub contains the merged sources and separate documentation-only policy references; Replit hosts a published TEST deployment, not certified production software. Real-world alert dispatch, legal clearance, WCAG conformance and deployed-security acceptance remain unverified. This report follows the organizer's six subject sections and five-page limit [1].